

Raise Alimak

03/10/2019

Applicability I

- *Alimak raise climber can drive long raise, vertical, inclines, curved and at any shape.
- *Alimak climber can be driven by air, electricity or diesel engine,
- *

Alimak Raise Climber I

- *Main mechanism for lifting Alimak raise climber is rail guide & penion.
- *Hoist climbs up by pin & rack mechanism.
- *Rack is part of guide rail & pins are over pully fitted on prime mover.
- *Guide rail is comprised of pipes for compressed air, water etc.

Alimak Raise Climber II

- *Guide rails are extended along with progress of raise.
- *Each guide rail is bolted to rock wall (hang wall side of raise) with help of expansion rock bolts.
- *Alimak raise climber has one man cage for carrying men & small tools.
- *Heavy materials are taken on drilling platform, above cage.

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Alimak Raise Climber III

- *Guide rail comes with different curvature, 8° , 25° etc.
- *Combination of curved guide make it possible to have raise as desired inclination.
- *Guide rail makes it possible to excavate very long raise vertical or incline.
- *Raise shape is mostly rectangular but curved shaped platform is available for excavating raise of any shape.
- *This method is suitable for blind raise.

Alimak Raise Climber IV

- *For very long raise, Alitrolly or Alicab is used for communicating in between lower level at raise face Alimak.
- *All work at raise face is performed from working platform, which is easily adjusted in height & angle.
- *Alimak manufacturer supplies platform of 1.6m X 1.6m, or 2.4m X 2.4m as standard, however other sizes are supplied on request.
- *Alimak platform has steel umbrella & fencing attached with platform for safety purpose.

Alimak Raise Climber V

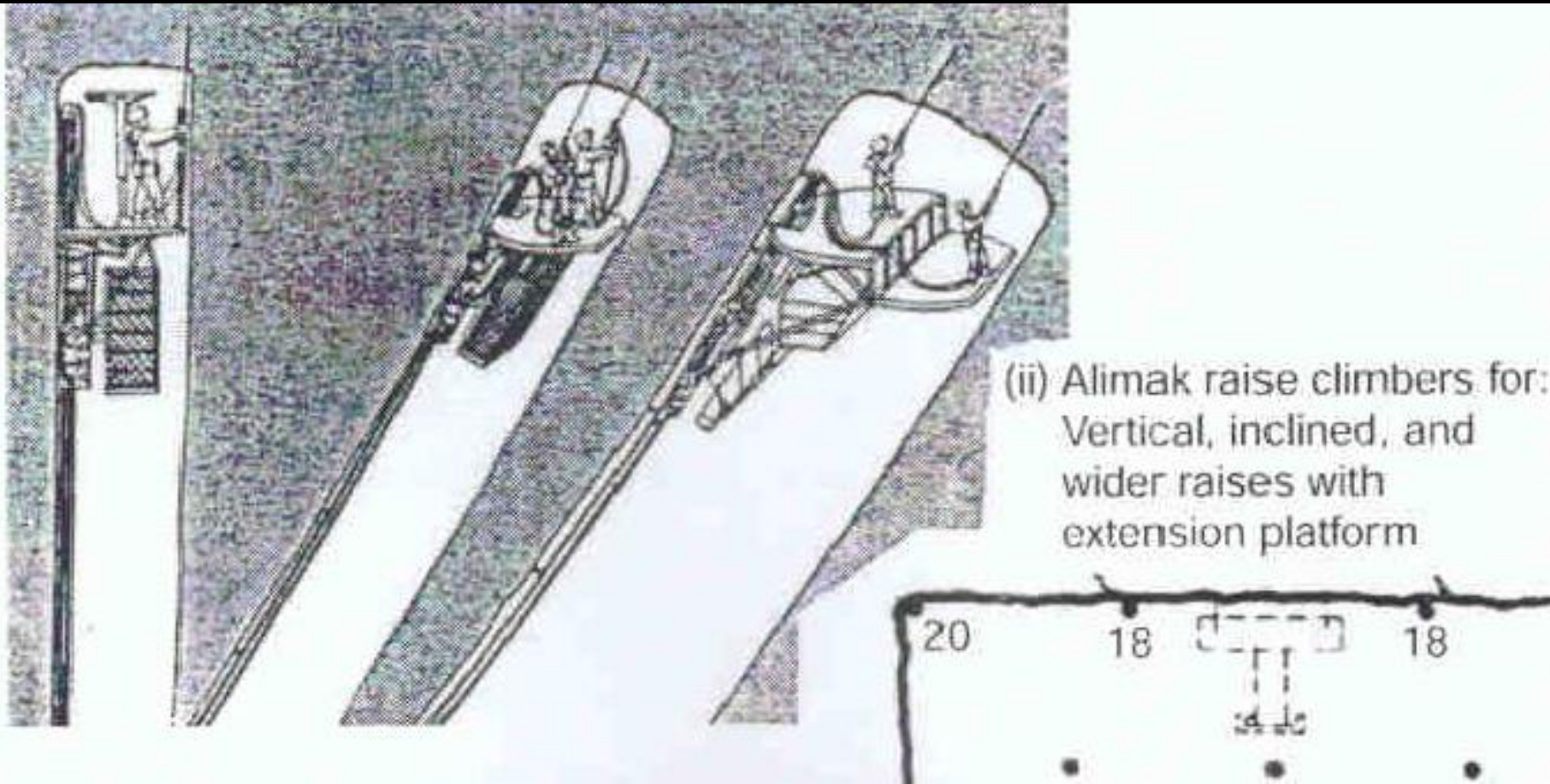
- *Two separate cable with distinct different cables placed through one pipe of guide rail.
- *One cable works as blasting cable & other cable works as telephone, control cable.
- *There are four pipe in guide rail, two for compressed air, one for water & fourth for communication, control, blasting & other cables.
- *These pipe line also provides extra rigidi to guide.

Alimak Raise Climber VI

- *Through control cable, water, air supply to header can be controlled.
- *For extension of guide rail, a guide rail of required length, bracket, expansion bolt, “O” ring, “U” washer, nuts & bolts, spacers, header plate with cocks are required.

Alimak Raise Vertical Incline I

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Alimak Raise I

*Guide Rail



Alimak Raise I

*We



Compressed Air Alimak Climber I

- *Alimak is driven with help of compressed air motor is common and mostly used in mines.
- *In compressed air operated Alimak raise climber, compressed air comes through hose & reel.
- *Weight of hose becomes & compressed air pressure are limiting factor for length of raise to be excavated.
- *Normal range of raise by compressed air machine is 200m but some raises up to 320m has been excavated.

Alimak Raise



Electric Alimak Climber I

- *Electric Alimak hoist gets power through electric cable.
- *Electric cables are specially designed to hang 1000m with out intermediate support.
- *Raising of about 1km can be done by electric Alimak raise climber.
- *Weight of 1000m cable varies from 1 to 1.6t
- *
- *

Electric Alimak Climber II

- *Cables are provided with extra tensile strength by incorporating steel wire & providing bondage between insulators & conductors.
- *This is done to arrest possible relative movement in between insulator & conductors.
- *Denoson Mines Canada has done 950m raising at 45° & 4sqm cross section.

Diesel Alimak Climber I

- *Diesel Hydraulic unit can excavate more than 1km or more long raise.
- *This unit is self-contained & does not require electric cable or compressed air hose.
- *For drilling, compressed air & water is made available through header (manifold).

Diesel Alimak Climber II

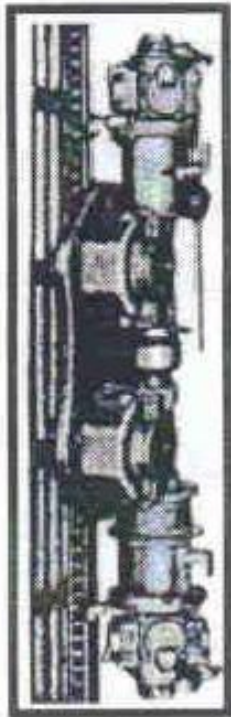
- *While ascending, diesel exhaust gas gets diluted by compresses air.
- *Lowering of Alimak raise is dine by gravity, without switching diesel unit.
- *While lowering Alimak, centrifugal brake controls speed of descending machine.

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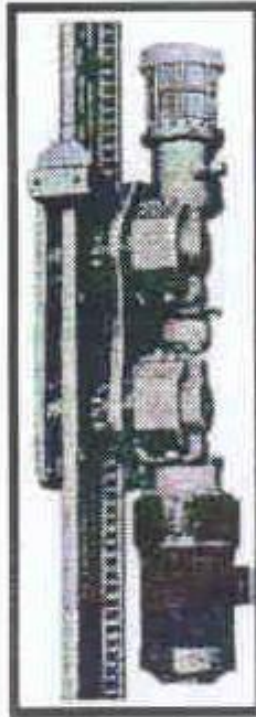
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Alimak Raise Traction I

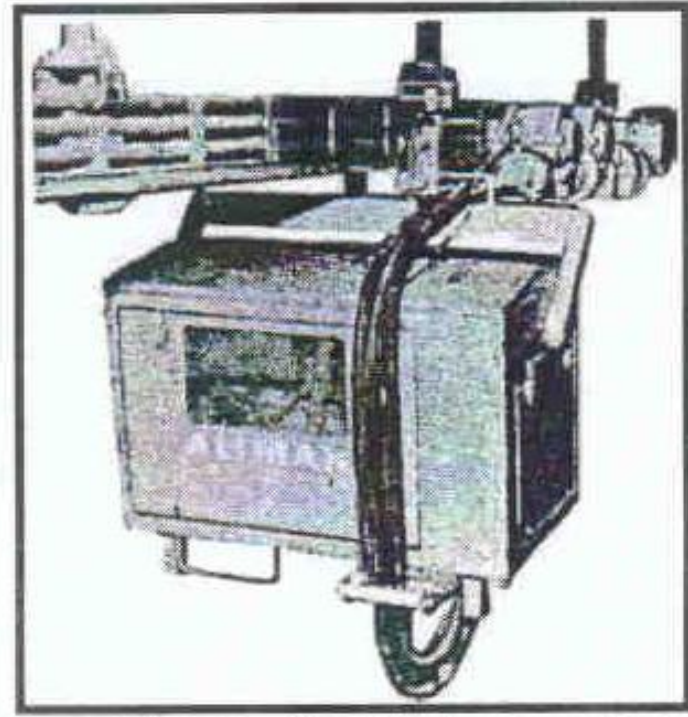
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Air



Electric



Diesel (i) Alimak raise climbers
motive power: Air, electric
and diesel

Preparation I

- *A horizontal cut out of 9 m (length) X 4 m (width) X 3 m (height) is excavated to accommodate raise climber with Alicab.
- *Without Alicab its length could be 7m.
- *For vertical raise curved guides are used 8, 25, 25, 25, 8 degree total 91 degree.
- *For installing Alimak raise climber at incline part, a 5 meter raise is excavated by open raise method.
- *For installing curved guide, brow is slashed at suitable angle.
- *Curved guide rail is assembled on ground and lifted with help of chain pulley block.

Preparation II

- *Guide curve is fixed by lifting it with help of pulley.
- *Size of platform is chosen based on the requirement of cross section of raise.
- *For larger platform size, Alimak raise drive should be designed accordingly.
- *A horizontal cut out (also known as raise access) of 9m (length) X 4m (width) x 3m (height) required to accommodate raise climber.
- *Without Alicab, length is reduced to 7m in place of 9m.

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Preparation III

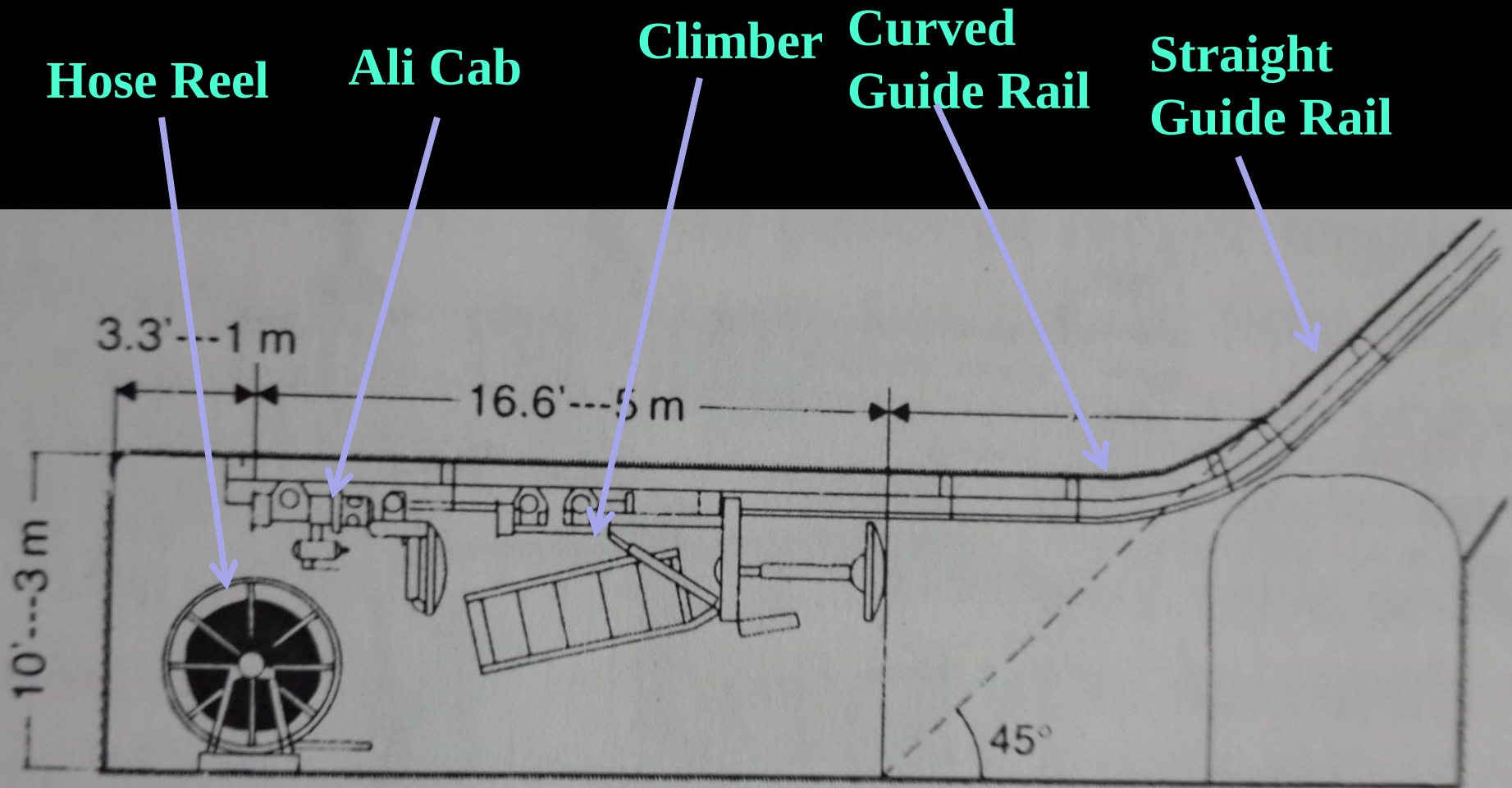
- *A vertical or incline raise has to be excavated by conventional method for a length of 5m.
- *Brow is slashed at 45 degree angle, 1m from corner.
- *Suitable curve guide rail is lifted with chain pully block & fixed.
- *

Preparation III++++

- *Guide rail of Alimak raise climber is anchored with rock wall with help of expansion bolt.
- *Expansion bolts are recoverable & can be used multiple times.
- *For compressed air operated Alimak raise climber, one compressed air connection and water connection is sufficient for raising.
- *Installation is simple and fast.
- *Electricity has the danger of electrocution.
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Alimak Raise I

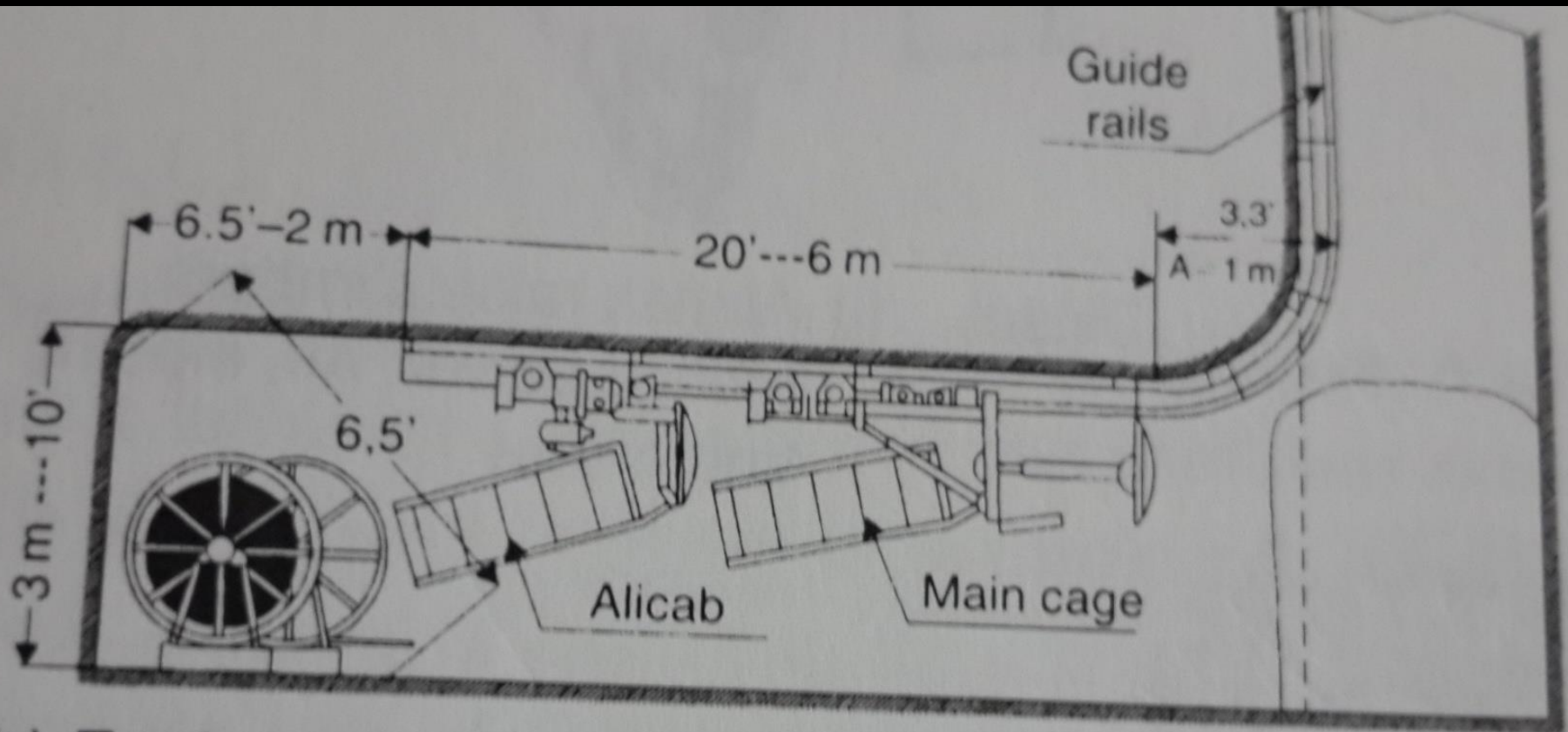
*Alimak for 45 degree raise.



(a) Equipment for an inclined raise

Alimak Raise I

*Alimak for vertical raise



(b) Equipment for vertical raise

Raising I

- *Drilling by Jack Hammer from working platform.
- *Normally burn cut is applied at raise face.
- *Burn is kept at foot wall of raise and angle of holes kept slightly steeper than raise.
- *Blast holes are charged with explosive from platform of Alimak.
- *Header is replaced with header plate.
- *Before blasting climber is brought to lower level & shaltered at place unaffected by blasting.

Raising II

- *After blasting air and water jets from header plate clear fumes and then face is approached under safety roof umbrella.
- *Face is scaled and after extension of guide rail, cycle is repeated.
- *After blasting compressed air & water mixture is sprayed at face, with out men being at raise face.
- *Drilling is done from working platform, on top of travelling cage.

Raising III

- *A safety belt must be used if gap (distance between side wall & platform) exceeds 15cm.
- *Drilling pattern is mostly burn cut, on some situation pyramid cut is also used.
- *While drilling holes, cut hole should be drilled in such a manner, so that through from cut hole does not hit guide rail.
- *On completion of drilling at face, holes are charged with explosive & header is replaced with header plate.

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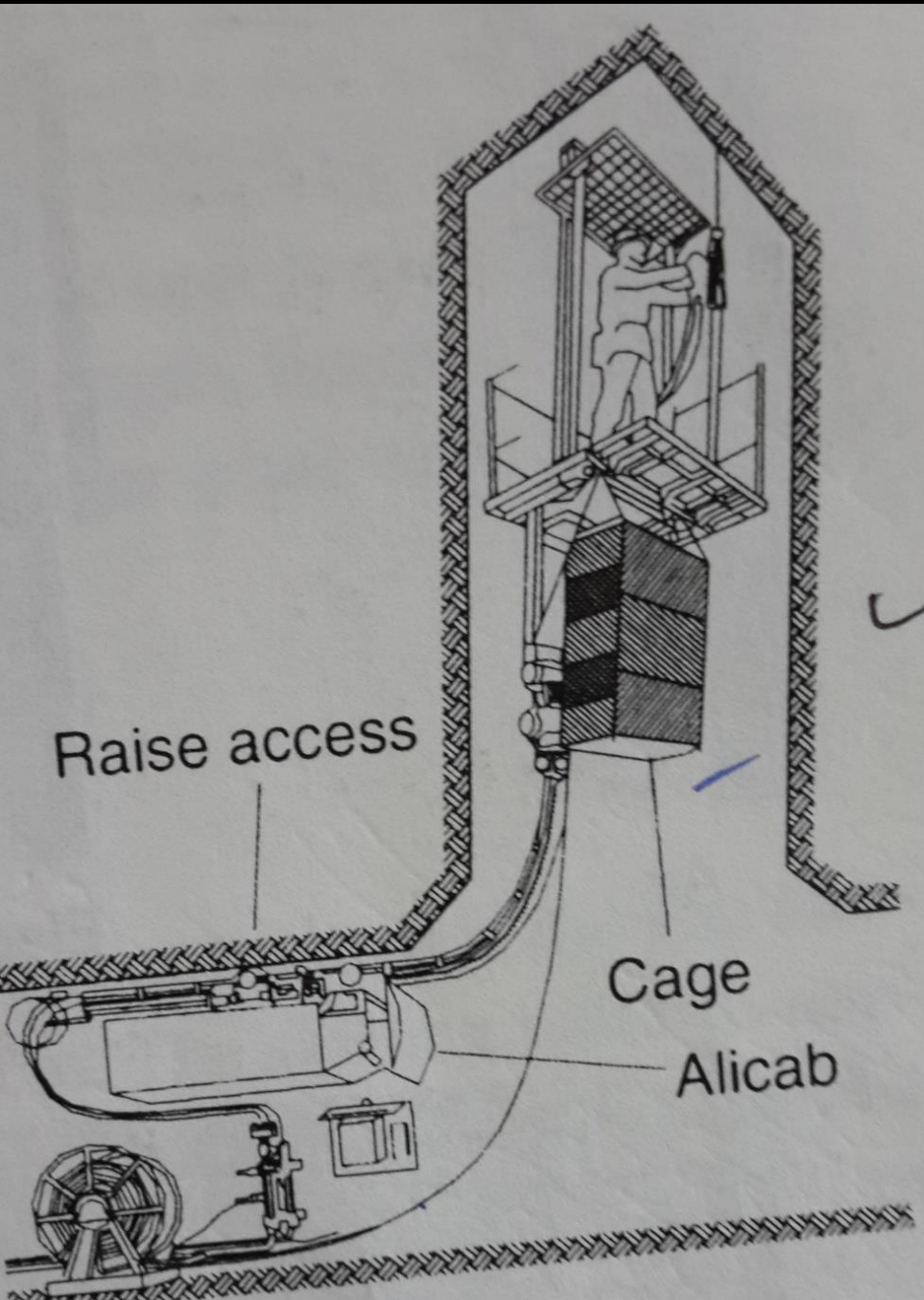
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Raising IV

- *Before blasting, climber is brought to raise inset.
- *After blasting, air & water jet from header plate clears fumes & face is scaled standing under safety roof (umbrella).
- *After scaling & extension of guide rail, cycle is repeated.
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- *

Alimak Raise I

*Alimak at face

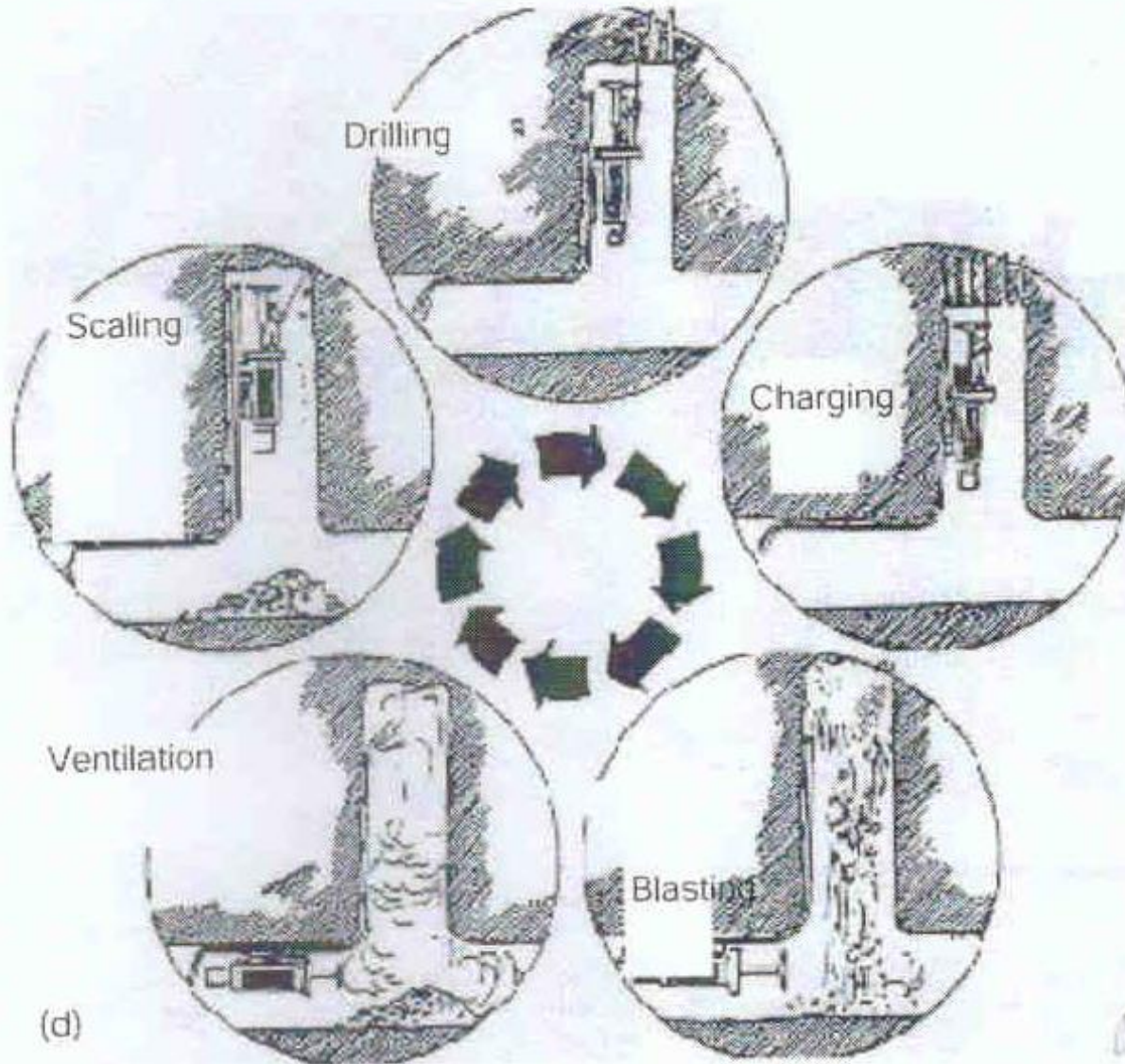


Alimak Raise



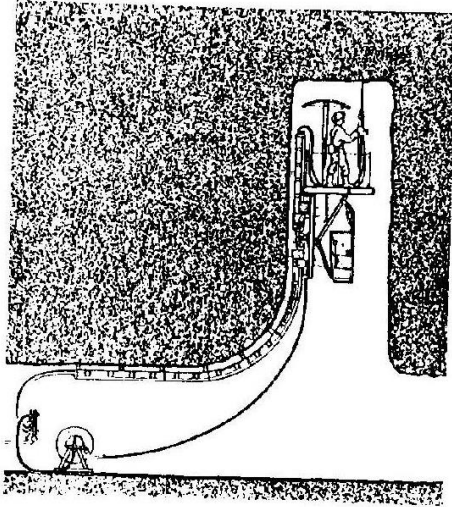
Alimak Raise Cycle I

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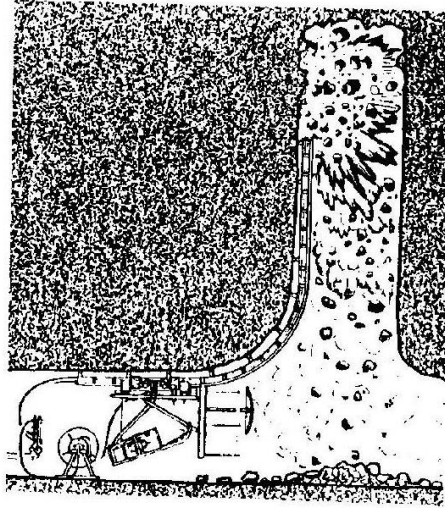


Raise Climber

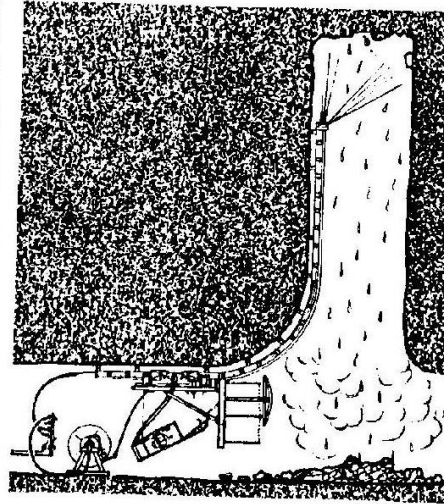
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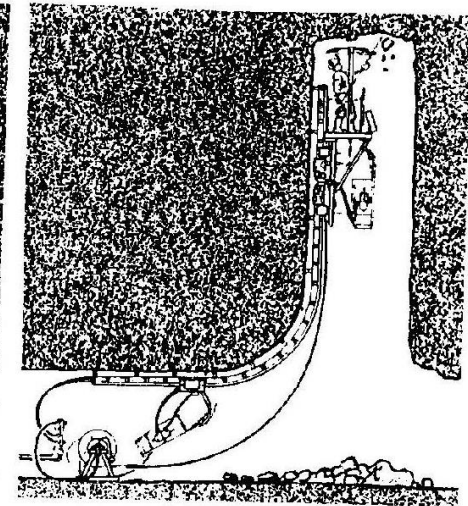
DRILLING: Scaling and extension of guide rail is safely and easily carried out from the platform under the steel protective roof. Tubes in the guide rail are used for the supply of air and water to the drills.



BLASTING: During blasting the Raise Climber is well protected in the tunnel.



CLEANING: After blasting the face is cleaned with an air and water mixture sprayed through jets in a plate protecting the top of the guide rail.



CLIMBING: The air motors are controlled from the cage, and they are supplied with air via a hose connected to a reeling drum in the tunnel. ALICAB or ALITROLLEY Reserve and Service Lifts provide fast communication with the Raise Climber and is useful for the transport of men and material, or in the case of accidents.

Blasting I

- *Blasting of face is done from lower level and from place of safety.
- *Blasting cable runs inside rail guide and remiss protected.

Alimak Raise Drill Hole Pattern I

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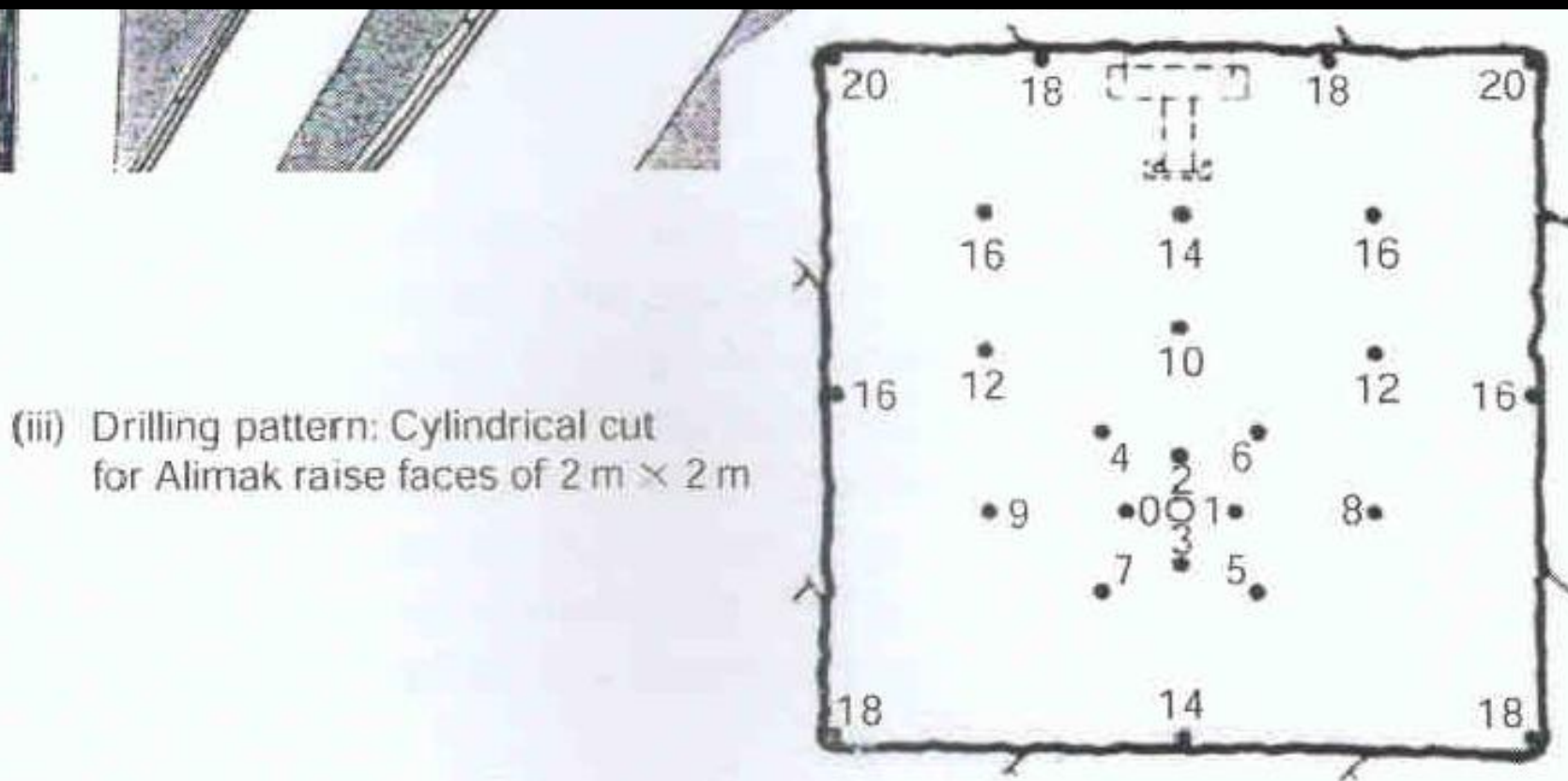


Figure 13.6 Alimak raise climber – Classification, applications and drilling pattern for 4 m^2 raise.

Ventilation I

*For ventilation raise face, one compressed air pipe and one water line are used available along with guide rail.

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Safety Features I

- *In absence of compressed air in drive motor, brake is automatically applied thus immobilizing the machines.
- *For bringing down Alimak in absence of air, brake has to be released manually.
- *Speed of descending Alimak with out power is controlled by centrifugal brake.
- *

Safety Features II

- *Efficient communication in between Alimak platform and level through telephone, bell etc.
- *Alicab can established physical communication with Alimak raise climber at any time.
- *Work platform can be easily adjusted with height of raise or cross section of raise.
- *Canopy above platform on Alimak raise climber reduces exposure of persons working on platform.

Safety Fetcher II

- *In case of non availability of traction power by prime mover, machine is immobilized automatically by a positive brake mechanism.
- *In case of non availability of power, machine can be lowered with help of gravity, by manually releasing brake.
- *In case of lowering machine with help of gravity, a centrifugal brake prevent machine from over speeding.

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Safety Featcher III

- *Man travels inside safety cage up to raise face, only comes out of cage at raise face.
- *Telephone communication with raise face is established through cable running through guide rail pipe.
- *Face is washed with water & air before man reaches at face.
- *This reduces chances of getting poisoned by blasting fume.
- *Man travels to freshly blasted face inside safety of cage & below working platform.

Safety Featcher I

- *Alicab / Alitrolly serves as emergency climber in system.
- *Gap between raise wall & Alimak platform is very small, chances of persons falling through this gap is eliminated.
- *If required, gap can be adjusted with suitable extension plate attached to platform.

Reference I

- *SME Mining Engineering Hand Book, 2nd Edition, Volume 2, Page 1622 to 1625.
- *Surface and Underground Excavation – Methods, Techniques and Equipment, Ratan Raj Tatiya. Page 336 to 340.